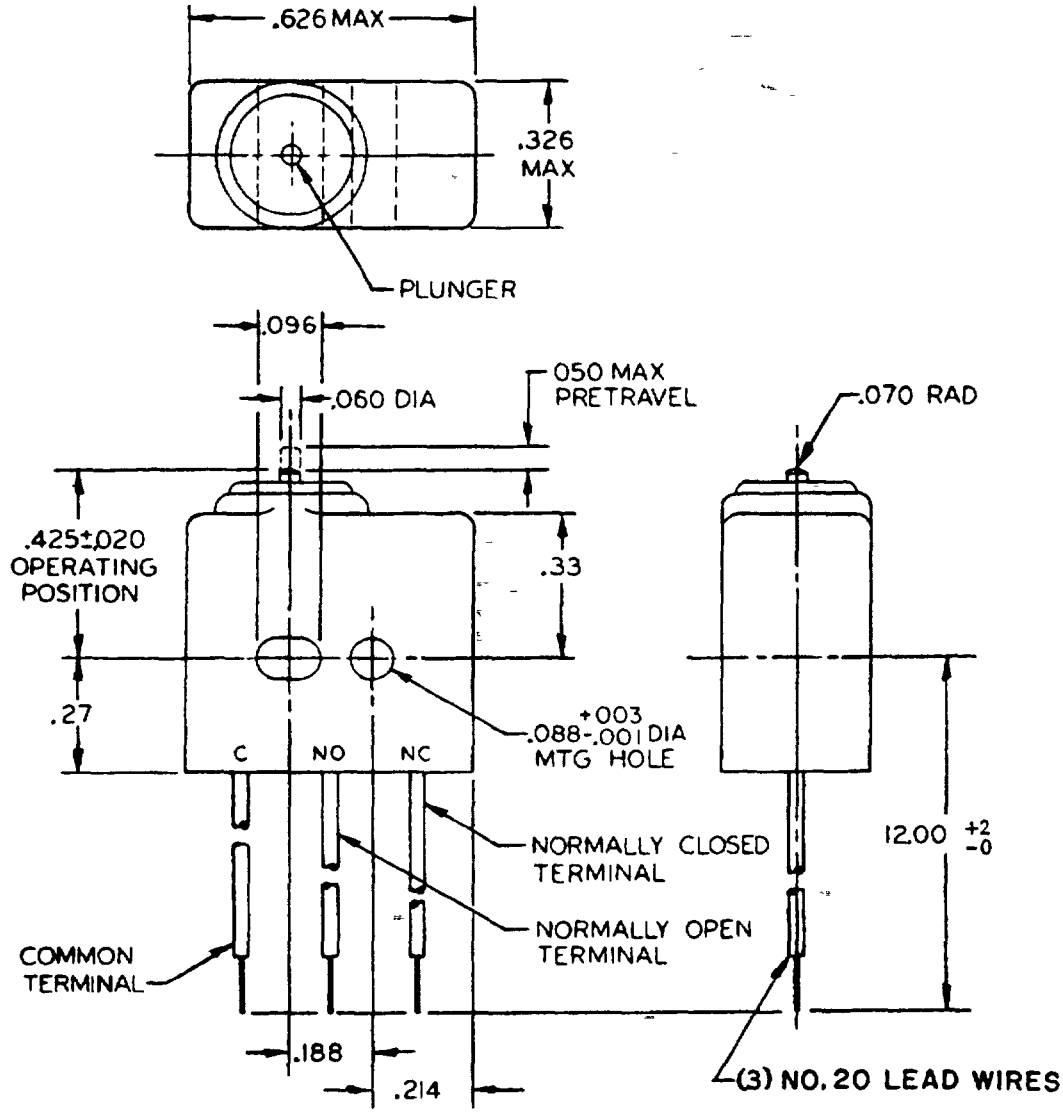
MILITARY SPECIFICATION SHEET

SWITCHES, SENSITIVE, SPDT, 7 AMPERES, WATERTIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-8805.



Wire lead terminals

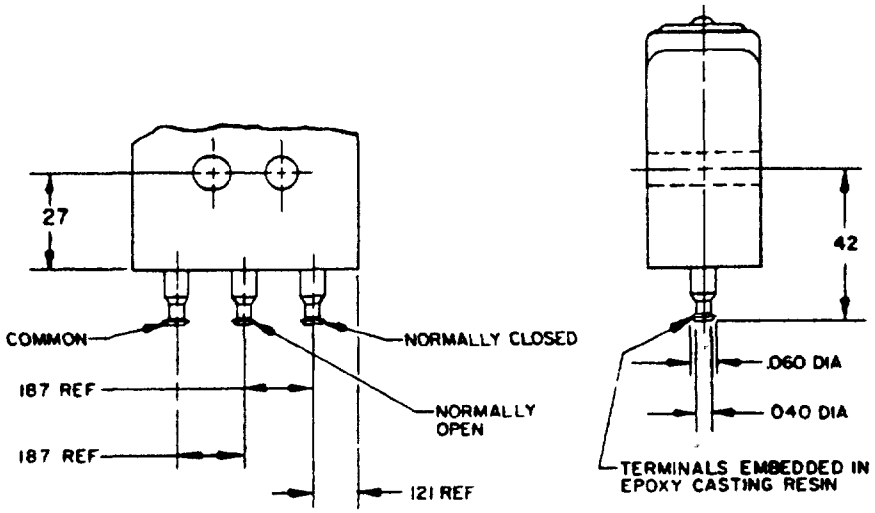
MS 27934-1,-2,-4 AND -5

FIGURE 1. Dimensions and configuration.

Ⓕ denotes changes

FSC 5930

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Turret terminals

MS 27994-3

INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.088	2.24	.326	8.28
.003	08	.096	2.44	.33	8.4
.020	.51	.121	3.07	.42	10.7
.040	1.02	.187	4.75	.425	10.80
.050	1.27	.188	4 78	.626	15.90
.060	1.52	.214	5.44	2.00	50.8
.070	1.78	.27	6.9	12.00	304.8

NOTES
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.02 (.5 mm) for two place decimals and ±.005 (.13 mm) for three place decimals.
3. Mounting holes will accept pins or screws of .087 (2 21 mm) maximum diameter on .188 ±.002 (4.78 ±.05 mm) centers.
4. Exact shape of switch optional, provided dimensions specified are not exceeded.
5 Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1 Dimensions and configuration - Continued

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REQUIREMENTS:

Dimensions and configuration: See figure I.

Enclosure design: 3 Watertight.

Temperature characteristic: 1 (-55°C to +85°C).

Shock type: M (100G).

Vibration grade: 2 (10 to 2,000 Hz).

Material: Type CFG in accordance with MIL-M-14 may be used internally.

Weight:

MS27994-1, -2, -4 and -5: .030 pound maximum.

MS27994-3: .01 pound maximum.

Operating characteristics:

Actuating force: 17 ounces maximum.

Release force: 4 ounces minimum.

Movement differential: .005 inch maximum.

Pretravel: .050 inch maximum.

Overtravel: .004 inch minimum.

Terminals:

Ⓕ MS27994-1 and -2: MIL-W-16878/4 wire leads.

MS27994-3: When terminals are gold plated, the finish shall be a minimum of .000030-inch and a maximum of .000100-inch thick.

Ⓕ MS27994-4 and -5: MIL-W-22759/11 wire leads.

Terminal strength:

MS27994-1, -2, -4 and -5: 20 pounds, wire leads.

MS27994-3: 9 pounds, solder terminals.

Contact resistance:

MS27994-1, -3 and -4: Not applicable.

MS27994-2 and -5: 70 milliohms, maximum.

Dielectric withstanding voltage:

At atmospheric pressure: 1,000 V rms.

Altitude: 400 V rms at 50,000 feet. In qualification inspection table after electrical endurance, the dielectric withstanding voltage points of application between all unconnected terminals of the same pole is not applicable.

Mechanical endurance: 25,000 cycles.

Electrical endurance: 25,000 cycles.

Electrical ratings: See table I.

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TABLE I. Electrical ratings.

Part number	Load	Sea level		50,000 ft
		28 V dc	115 V ac, 400 Hz	28 V dc
		(Amperes)	(Amperes)	(Amperes)
MS27994-1. -3, -4 1/ (Silver contacts)	Resistive	7	7	7
	Inductive	4	4	2.5
	Motor	2.5	1.5	---
	Lamp	1	1	---
MS27994-2 and -5 1/ (Gold contacts)	Resistive	1	---	1
	Inductive	.5	---	.5

Ⓕ 1/ MS27994-1 and -2 are inactive for new design.

Qualification.

Group submission.

Sample: The number of sample units for each part number covered by this specification sheet shall be as specified in table II.

Test routine. The sample units shall be divided as specified in table II and subjected to the inspections indicated.

TABLE II. Qualification inspection.

Switches for testing	Number of sample units	Examination or test	Extent of approval
MS27994-4	30	Complete qualification tables, as applicable, of MIL-S-8805	MS27994-1 and -4
MS27994-5	8	Group I Visual and mechanical examination Insulation resistance Operating characteristics	
	8 from group I	Group II Overload cycling Electrical endurance 1/ At sea level: Resistive load, dc Inductive load, dc At reduced barometric pressure: Resistive load, dc Inductive load, dc Dielectric withstanding voltage 2/ Operating characteristics (All group II sample units)	MS27994-2 and -5 3/

See footnotes at end of table.

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TABLE II. Qualification inspection - Continued.

Switches for testing	Number of sample units	Examination or test	Extent of approval
MS27994-3	2	Group I Visual and mechanical examination Insulation resistance Operating characteristics Terminal strength Dielectric withstanding voltage	MS27994-3 3/

1/ Two sample units for each electrical load.

2/ Sea level dielectric withstanding voltage test is to be conducted only on those units which were tested for sea level electrical endurance. Altitude dielectric withstanding voltage test is to be conducted on those units which were tested for altitude electrical endurance tests.

Ⓕ 3/ Applies only if MS27994-4 is qualified.

Part number: See table III.

TABLE III. Part number.

Part number	Contact	Termination
MS27994-1	Silver	Wire leads 1/
MS27994-2	Gold, filled or diffused	Wire leads 2/
MS27994-3	Silver	Turret terminals
MS27994-4	Silver	Wire leads 3/
MS27994-5	Gold, filled or diffused	Wire leads 3/

Ⓕ 1/ MIL-W-16878/4 wire leads. Inactive for new design. Use MS27994-4.

2/ MIL-W-16878/4 wire leads. Inactive for new design. Use MS27994-5.

3/ MIL-W-22759/11 wire leads.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy -AS, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - MI, AV
Navy -MC, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5930-1204-1)

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